

How to automate result reporting

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<https://github.com/jimb0w/automation>

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1 Data cleaning

Document everything! Say what you're going to do, do it, then comment on it.

For example: I'm going to create a random health dataset with the following variables:

- Sex (0 = female; 1 = male)
- Age (in years)
- Diabetes status (0 = no diabetes; 1 = diabetes)
- Smoking status (0 = non-smoker; 1 = smoker)
- LDL-C (in mmol/L)
- Outcome (in outcome units)

As so:

```
cd /home/jimb0w/Downloads/tempauto/  
clear  
set obs 20000  
set seed 1312  
gen sex = runiformint(0,1)  
gen age = rnormal(60,15)  
gen diabetes = runiformint(0,1)  
gen smoking = runiformint(0,1)  
gen ldl = rnormal(3,0.5)  
gen outcome = rnormal(200,30)+diabetes*rnormal(100,20)  
export delimited using randomhealthdata.csv, replace  
save randomhealthdata, replace
```

Now I'm going to check the variables make sense:

```
. su(sex), detail
```

sex				
	Percentiles	Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	20,000
25%	0	0	Sum of wgt.	20,000
50%	0		Mean	.4989
		Largest	Std. dev.	.5000113
75%	1	1		
90%	1	1	Variance	.2500113
95%	1	1	Skewness	.0044
99%	1	1	Kurtosis	1.000019

```
. su(diab), detail
```

diabetes				
	Percentiles	Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	20,000
25%	0	0	Sum of wgt.	20,000
50%	0		Mean	.494
		Largest	Std. dev.	.4999765
75%	1	1		
90%	1	1	Variance	.2499765
95%	1	1	Skewness	.0240017
99%	1	1	Kurtosis	1.000576

```
. su(smoking), detail
```

smoking				
	Percentiles	Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	20,000
25%	0	0	Sum of wgt.	20,000
50%	1		Mean	.50085
		Largest	Std. dev.	.5000118
75%	1	1		
90%	1	1	Variance	.2500118
95%	1	1	Skewness	-.0034
99%	1	1	Kurtosis	1.000012

```
. su(age), detail
```

age				
	Percentiles	Smallest		
1%	24.66596	.835979		
5%	35.15182	1.354185		
10%	40.73132	4.960024	Obs	20,000
25%	49.76038	5.434331	Sum of wgt.	20,000
50%	59.77124		Mean	59.82388
		Largest	Std. dev.	14.99574
75%	69.96964	109.9906		
90%	78.95889	111.0537	Variance	224.8723
95%	84.55445	111.059	Skewness	-.0096071
99%	94.6009	115.5846	Kurtosis	3.032043

```
. su(ldl), detail
```

ldl				
	Percentiles	Smallest		
1%	1.81618	1.169115		
5%	2.166875	1.17411		
10%	2.354322	1.184871	Obs	20,000
25%	2.66082	1.20128	Sum of wgt.	20,000
50%	3.000187		Mean	2.997539
		Largest	Std. dev.	.50188
75%	3.334033	4.78299		
90%	3.636441	4.849288	Variance	.2518836
95%	3.817179	4.876576	Skewness	-.0133279
99%	4.168352	5.034541	Kurtosis	3.031805

```
. su(outcome), detail
```

outcome				
	Percentiles	Smallest		
1%	139.2732	87.21138		
5%	162.0865	91.83277		
10%	174.7943	91.93818	Obs	20,000
25%	199.8294	92.32668	Sum of wgt.	20,000
50%	244.0778		Mean	249.3994
		Largest	Std. dev.	59.80981
75%	299.1514	426.0753		
90%	329.6298	430.0988	Variance	3577.214
95%	345.9959	434.5064	Skewness	.163877
99%	374.6187	450.9283	Kurtosis	2.073516

It looks like these variables make sense and there don't appear to be any errors with the data. But we should check further: let's produce a histogram to check the continuous variables. The code below produces Figure 1.1.

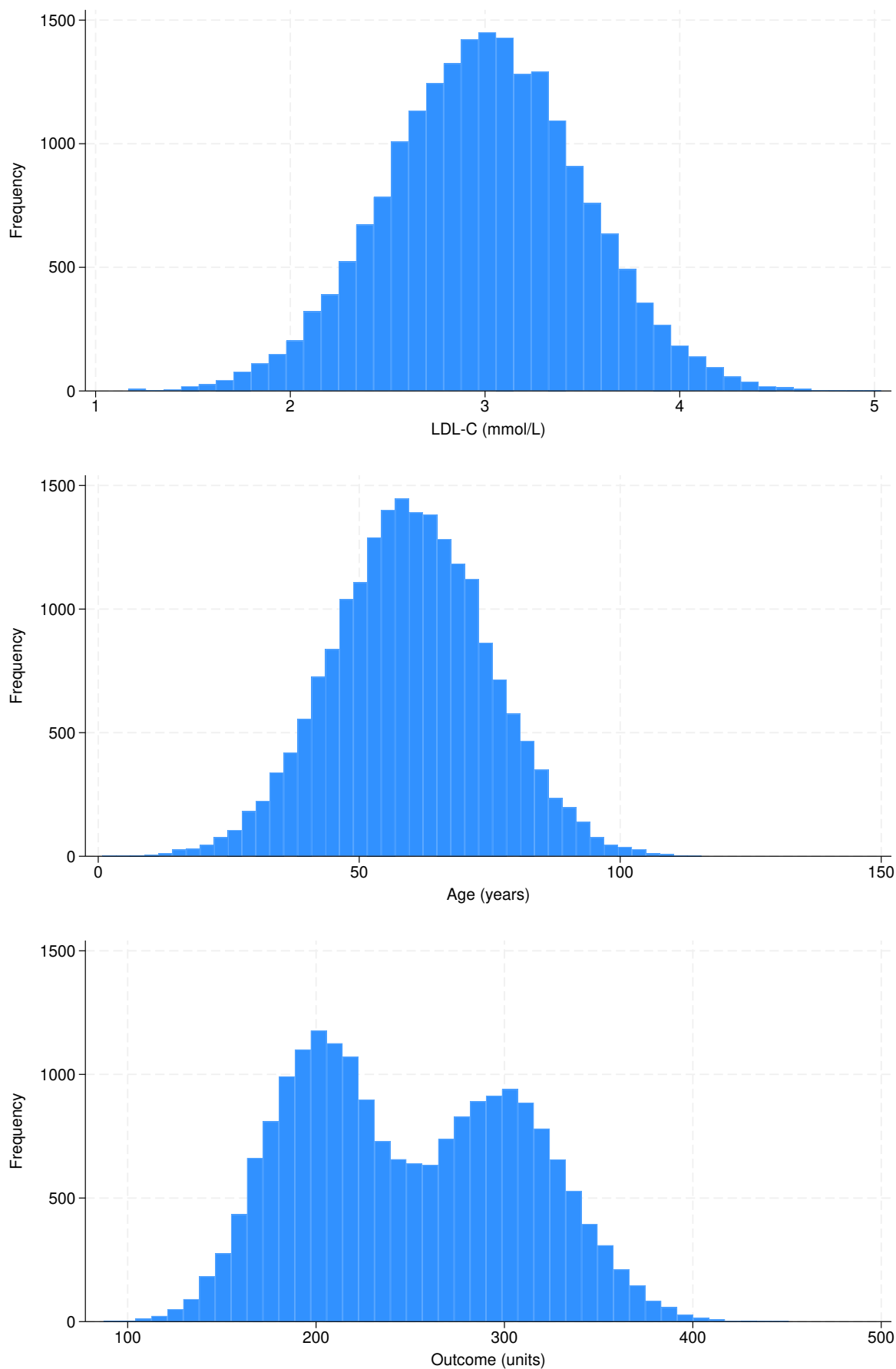
```
mkdir GPH
```

```

hist ldl, frequency ///
xtitle("LDL-C (mmol/L)")
graph save GPH/LDL_hist, replace
hist age, frequency ///
xtitle("Age (years)")
graph save GPH/Age_hist, replace
hist outcome, frequency ///
xtitle("Outcome (units)")
graph save GPH/Outcome_hist, replace
graph combine ///
GPH/LDL_hist.gph ///
GPH/Age_hist.gph ///
GPH/Outcome_hist.gph ///
, cols(1) altshrink xsize(3)
graph export GPH/distcheck.pdf, as(pdf) replace

```

Figure 1.1: Histograms of LDL-C, Age, and outcome.



It appears that our age and LDL-C variables are normally distributed and seem reasonable. there is an odd, bimodal distribution in the outcome variable.

In terms of automation, the key thing here was producing my Figures using code, rather than a graphics program, and automatically exporting them to pdf, before importing them and displaying them within the document. Notice also that Figure 1.1 hyperlinks to the figure if referenced properly, so if you update the figure order later, you don't have to re-number all your figure references in the text. That means if you change anything in your analysis, add a figure, etc., you don't need to re-make the figure, just re-run the code and it will re-appear updated in this pdf file.

2 Analysis

Before beginning our analysis, we should present a summary table of population characteristics. We are most interested in diabetes, so we intend to stratify this table by diabetes type.

Here, we are going to collect the results from the Stata output and store them in a matrix, before saving that matrix to the Stata dataset editor and formatting the outputs for presentation in a Table. It helps to first think about the Table you want to present before collecting the results. The Table 2.1 is what we're aiming for.

Table 2.1: Table we are aiming for.

Overall		Diabetes status	
		No diabetes	Diabetes
N	N	N	N
Females	N (%)	N (%)	N (%)
Smoking status	N (%)	N (%)	N (%)
Age	median (IQR)	median (IQR)	median (IQR)
LDL-C (mmol/L)	median (IQR)	median (IQR)	median (IQR)
Outcome	median (IQR)	median (IQR)	median (IQR)

Data are presented as N, N (% of column total), or median (IQR).

Now that we know what we are aiming for, we need to collect the values. Stata's *return list* command is useful here: it tells you what is stored in Stata's memory, and thus how we can access it. For example:

```
. use randomhealthdata, clear
. su(ldl), detail
```

ldl				
	Percentiles	Smallest		
1%	1.81618	1.169115		
5%	2.166875	1.17411		
10%	2.354322	1.184871	Obs	20,000
25%	2.66082	1.20128	Sum of wgt.	20,000
50%	3.000187		Mean	2.997539
		Largest	Std. dev.	.50188
75%	3.334033	4.78299		
90%	3.636441	4.849288	Variance	.2518836
95%	3.817179	4.876576	Skewness	-.0133279
99%	4.168352	5.034541	Kurtosis	3.031805

```
. return list
scalars:
      r(N) = 20000
    r(sum_w) = 20000
    r(mean) = 2.99753932904005
      r(Var) = .2518835571674297
      r(sd) = .5018800226821443
    r(skewness) = -.0133279276969729
    r(kurtosis) = 3.031805267881794
      r(sum) = 59950.78658080101
      r(min) = 1.169115424156189
      r(max) = 5.03454065322876
      r(p1) = 1.816179573535919
      r(p5) = 2.166874527931213
      r(p10) = 2.354321837425232
      r(p25) = 2.660820245742798
```

```

r(p50) = 3.000186562538147
r(p75) = 3.334033250808716
r(p90) = 3.636441469192505
r(p95) = 3.817179083824158
r(p99) = 4.168352127075195

```

We can access any one of those statistics and store it for use in our table. Annoyingly *tabulate* doesn't store anything useful, so you have to use the *matcell* option.

So, let's get all the data we need for our table:

```

count
local a = r(N)
count if sex == 0
local b = r(N)
count if smoking == 1
local c = r(N)
su(age), detail
mat A1 = (r(p50),r(p25),r(p75))
su(ldl), detail
mat A2 = (r(p50),r(p25),r(p75))
su(outcome), detail
mat A3 = (r(p50),r(p25),r(p75))
mat B = (0`a`,... ///
        \0`b`,... ///
        \0`c`,... ///
        \A1\A2\A3)
forval i = 0/1 {
count if diabetes == `i`
local a = r(N)
count if sex == 0 & diabetes == `i`
local b = r(N)
count if smoking == 1 & diabetes == `i`
local c = r(N)
su(age) if diabetes == `i`, detail
mat A1 = (r(p50),r(p25),r(p75))
su(ldl) if diabetes == `i`, detail
mat A2 = (r(p50),r(p25),r(p75))
su(outcome) if diabetes == `i`, detail
mat A3 = (r(p50),r(p25),r(p75))
mat B`i` = (0`a`,... ///
           \0`b`,... ///
           \0`c`,... ///
           \A1\A2\A3)
}
mat C = (B,B0,B1)

. mat l C
C[6,9]

```

	c1	c2	c3	c1	c2	c3	c1	c2
r1	20000	.	.	10120	.	.	9880	.
r2	10022	.	.	5036	.	.	4986	.
r3	10017	.	.	5061	.	.	4956	.
r1	59.771242	49.76038	69.969643	59.866039	49.805601	70.10405	59.671923	49.716722
r1	3.0001866	2.6608202	3.3340333	2.9967493	2.6569303	3.3266593	3.0038455	2.6666859
r1	244.0778	199.82938	299.1514	200.40833	180.11034	220.09869	299.5594	275.5574

```

c3
r1 .
r2 .
r3 .
r1 69.824024
r1 3.3387761
r1 323.75455

```

We have all our results in the matrix, *C*. However, they aren't in a presentable format. If we were to try and make a table out of this, we would have too many decimal places and inconsistent

column numbers. So, we need to convert the variables to string (essentially text, not number) format so that we can manipulate the data into the format we need for our table.

```
mkdir CSV
clear
svmat C
gen D1 = C1[1]
gen D4 = C4[1]
gen D7 = C7[1]
gen E1 = 100*C1/D1
gen E4 = 100*C4/D4
gen E7 = 100*C7/D7
tostring C1 C4 C7, gen(A1 A4 A7) force format(%9.0fc)
tostring _all, replace force format(%9.1f)
gen A = A1 if _n == 1
gen B = A4 if _n == 1
gen C = A7 if _n == 1
replace A = A1 + " (" + E1 + "%)" if inrange(_n,2,3)
replace B = A4 + " (" + E4 + "%)" if inrange(_n,2,3)
replace C = A7 + " (" + E7 + "%)" if inrange(_n,2,3)
replace A = C1 + " (" + C2 + ", " + C3 + ")" if _n >=4
replace B = C4 + " (" + C5 + ", " + C6 + ")" if _n >=4
replace C = C7 + " (" + C8 + ", " + C9 + ")" if _n >=4
keep A B C
gen row = ""
replace row = "N" if _n == 1
replace row = "Females" if _n == 2
replace row = "Smoking status" if _n == 3
replace row = "Age" if _n == 4
replace row = "LDL-C (mmol/L)" if _n == 5
replace row = "Outcome" if _n == 6
order row
export delimited using CSV/table1.csv, delimiter(":") novarnames replace
save table1, replace
```

We have successfully produced our first table. I like to save tables in a CSV subdirectory to make the main folder cleaner. That also makes it easier to bulk save all your results to GitHub.

There are several ways to use tables in a LaTeX document. I use *pgfplotstable* as it allows me to export the file to a csv document and then use LaTeX. You could also simply write a TeX table in Stata and export that.

Table 2.2: Population characteristics.

	Overall	Diabetes status	
		No diabetes	Diabetes
N	20,000	10,120	9,880
Females	10,022 (50.1%)	5,036 (49.8%)	4,986 (50.5%)
Smoking status	10,017 (50.1%)	5,061 (50.0%)	4,956 (50.2%)
Age	59.8 (49.8, 70.0)	59.9 (49.8, 70.1)	59.7 (49.7, 69.8)
LDL-C (mmol/L)	3.0 (2.7, 3.3)	3.0 (2.7, 3.3)	3.0 (2.7, 3.3)
Outcome	244.1 (199.8, 299.2)	200.4 (180.1, 220.1)	299.6 (275.6, 323.8)

We can see from the table that the Outcome seems to be higher in people with diabetes (median (IQR): 299.6 (275.6, 323.8)) than without diabetes (median (IQR): 200.4 (180.1, 220.1)).

Okay, now it's your turn. Use these tools to run a linear regression analysis comparing each explanatory variable with the outcome, show the results in a table and a figure that is completely automatic. Then comment on the results in the text using results generated automatically.